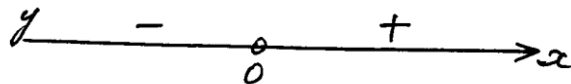


# Практическое занятие 14

## Построение графиков функций. Построение линий в полярных координатах

1) Построить график функции

$$y = x e^{\frac{1}{x}}, \quad D(y) = (-\infty; 0); (0; +\infty)$$



$$\lim_{x \rightarrow 0-0} y = \lim_{x \rightarrow 0-0} x e^{\frac{1}{x}} = 0$$

$$\begin{aligned} \lim_{x \rightarrow 0+0} y &= \lim_{x \rightarrow 0+0} x e^{\frac{1}{x}} = (0 * \infty) = \lim_{x \rightarrow 0+0} \frac{e^{\frac{1}{x}}}{\frac{1}{x}} = \left[ t = \frac{1}{x}, t \rightarrow +\infty \right] \\ &= \lim_{t \rightarrow +\infty} \frac{e^t}{t} = \left( \frac{\infty}{\infty} \right) = \lim_{t \rightarrow +\infty} \frac{e^t}{1} = +\infty \end{aligned}$$

$x=0$  – вертикальная асимптота

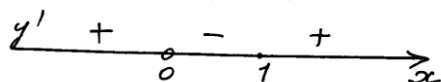
$$\kappa = \lim_{x \rightarrow \infty} \frac{y}{x} = \lim_{x \rightarrow \infty} e^{\frac{1}{x}} = e^0 = 1$$

$$\sigma = \lim_{x \rightarrow \infty} (y - \kappa x) = \lim_{x \rightarrow \infty} \left( x e^{\frac{1}{x}} - x \right) = \lim_{x \rightarrow \infty} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} = \left[ t = \frac{1}{x}, t \rightarrow 0 \right] =$$

$$\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = \left( \frac{0}{0} \right) = \lim_{t \rightarrow 0} \frac{t}{t} = 1$$

$y=x+1$  – наклонная асимптота

$$y' = e^{\frac{1}{x}} + x \left( -\frac{1}{x^2} \right) e^{\frac{1}{x}} = e^{\frac{1}{x}} - \frac{1}{x} e^{\frac{1}{x}} = \frac{x-1}{x} e^{\frac{1}{x}}$$



$$y' = 0, \quad \text{если } x = 1$$

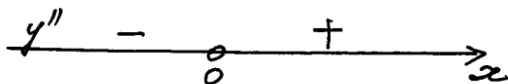
$x = 1$  – точка минимума

$$y(1) = e$$

$$\lim_{x \rightarrow 0-0} y' = \lim_{x \rightarrow 0-0} \left( e^{\frac{1}{x}} - \frac{1}{x} e^{\frac{1}{x}} \right) = 0$$

$$\begin{aligned} \lim_{x \rightarrow 0-0} \frac{1}{x} e^{\frac{1}{x}} &= (\infty * 0) = \left[ \frac{1}{x} = -t, t \rightarrow +\infty \right] = \lim_{t \rightarrow +\infty} (-t e^{-t}) \\ &= \lim_{t \rightarrow +\infty} \left( -\frac{t}{e^t} \right) = 0 \end{aligned}$$

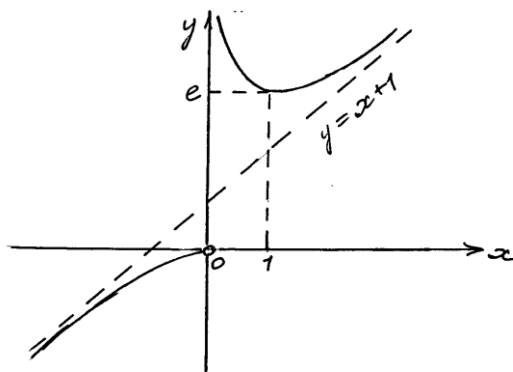
$$y'' = -\frac{1}{x^2} e^{\frac{1}{x}} + \frac{1}{x^2} e^{\frac{1}{x}} + \frac{1}{x^3} e^{\frac{1}{x}} = \frac{1}{x^3} e^{\frac{1}{x}}$$



Перегибов нет.

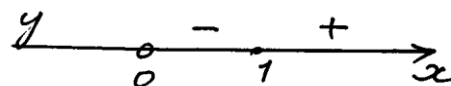
|     |                |     |          |     |                |
|-----|----------------|-----|----------|-----|----------------|
| $x$ | $(-\infty; 0)$ | $0$ | $(0; 1)$ | $1$ | $(1; +\infty)$ |
|-----|----------------|-----|----------|-----|----------------|

|       |                                                                                   |             |                                                                                    |               |                                                                                     |
|-------|-----------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------|
| $y'$  | +                                                                                 | $\nexists$  | -                                                                                  | 0             | +                                                                                   |
| $y''$ | -                                                                                 | $\nexists$  | +                                                                                  |               | +                                                                                   |
| $y$   |  | $x=0$ (в а) |  | $y_{min} = e$ |  |



2) Построить график функции

$$y = x^2 \ln x, \quad D(y) = (0; +\infty)$$



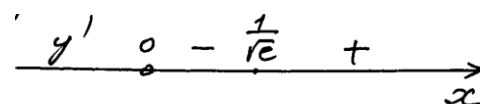
$$\begin{aligned} \lim_{x \rightarrow 0+0} y &= \lim_{x \rightarrow 0+0} x^2 \ln x = (0 * \infty) = \lim_{x \rightarrow 0+0} \frac{\ln x}{\frac{1}{x^2}} = \left( \frac{\infty}{\infty} \right) = \lim_{x \rightarrow 0+0} \frac{\frac{1}{x}}{-\frac{2}{x^3}} \\ &= \lim_{x \rightarrow 0+0} \left( -\frac{x^2}{2} \right) = 0 \end{aligned}$$

Вертикальных асимптот нет

$$\kappa = \lim_{x \rightarrow +\infty} \frac{y}{x} = \lim_{x \rightarrow +\infty} (x \ln x) = \infty$$

Наклонной асимптоты нет.

$$y' = 2x \ln x + x = x(2 \ln x + 1)$$



$$y' = 0, \text{ если } 2 \ln x + 1 = 0$$

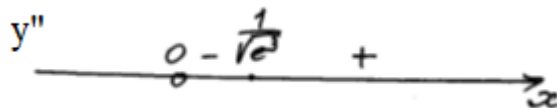
$$x = \frac{1}{\sqrt{e}} - \text{точка минимума}$$

$$\ln x = -\frac{1}{2}, x = e^{\frac{1}{2}}$$

$$y\left(\frac{1}{\sqrt{e}}\right) = -\frac{1}{2e}$$

$$\lim_{x \rightarrow 0+0} y' = \lim_{x \rightarrow 0+0} \frac{2 \ln x + 1}{\frac{1}{x}} = \left( \frac{\infty}{\infty} \right) = \lim_{x \rightarrow 0+0} \frac{\frac{2}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0+0} (-2x) = 0$$

$$y'' = 2 \ln x + 2 + 1 = 2 \ln x + 3$$



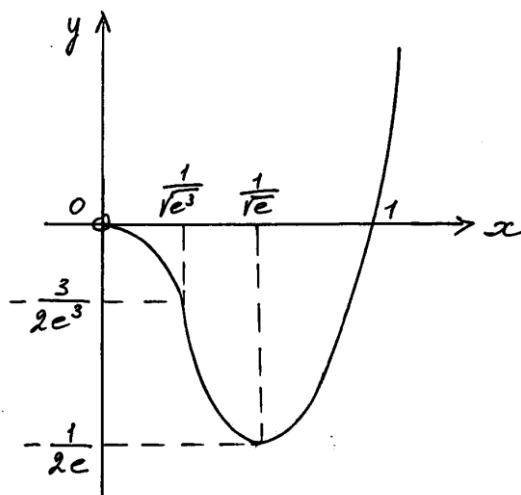
$$y'' = 0, \text{ если } 2 \ln x + 3 = 0$$

$$x = e^{-\frac{3}{2}} - \text{точка перегиба}$$

$$\ln x = -\frac{3}{2} \quad x = e^{-\frac{3}{2}}$$

$$y\left(\frac{1}{\sqrt{e^3}}\right) = -\frac{3}{2e^3}$$

|       |                                   |                                        |                                    |                                                         |                                  |                                            |
|-------|-----------------------------------|----------------------------------------|------------------------------------|---------------------------------------------------------|----------------------------------|--------------------------------------------|
| $x$   | $0$                               | $\left(0; \frac{1}{\sqrt{e^3}}\right)$ | $\frac{1}{\sqrt{e^3}}$             | $\left(\frac{1}{\sqrt{e^3}}; \frac{1}{\sqrt{e}}\right)$ | $\frac{1}{\sqrt{e}}$             | $\left(\frac{1}{\sqrt{e}}; +\infty\right)$ |
| $y'$  | $\lim_{x \rightarrow 0+0} y' = 0$ | -                                      |                                    | -                                                       | $0$                              | +                                          |
| $y''$ |                                   | -                                      | $0$                                | +                                                       |                                  | +                                          |
| $y$   | $\lim_{x \rightarrow 0+0} y = 0$  |                                        | $y_{\text{пер}} = -\frac{3}{2e^3}$ |                                                         | $y_{\text{min}} = -\frac{1}{2e}$ |                                            |



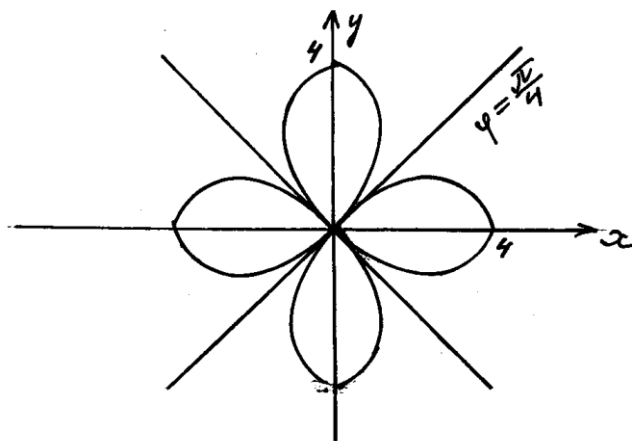
3) Построить линию, заданную уравнением в полярных координатах

$$r = 4 \cos^2 2\varphi$$

$$r = 2(1 + \cos 4\varphi), \quad T = \frac{2\pi}{4} = \frac{\pi}{2}$$

$r(-\varphi) = r(\varphi)$  – симметрия относительно полярной оси

|                 |     |                      |                      |                  |                 |                 |                       |                       |                 |
|-----------------|-----|----------------------|----------------------|------------------|-----------------|-----------------|-----------------------|-----------------------|-----------------|
| $\varphi$       | $0$ | $\frac{\pi}{24}$     | $\frac{\pi}{16}$     | $\frac{\pi}{12}$ | $\frac{\pi}{8}$ | $\frac{\pi}{6}$ | $\frac{3\pi}{16}$     | $\frac{5\pi}{24}$     | $\frac{\pi}{4}$ |
| $\cos 4\varphi$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$    | $0$             | $-\frac{1}{2}$  | $-\frac{\sqrt{2}}{2}$ | $-\frac{\sqrt{3}}{2}$ | $-1$            |
| $r$             | $4$ | $2+\sqrt{3}$         | $2+\sqrt{2}$         | $3$              | $2$             | $1$             | $2-\sqrt{2}$          | $2-\sqrt{3}$          | $0$             |

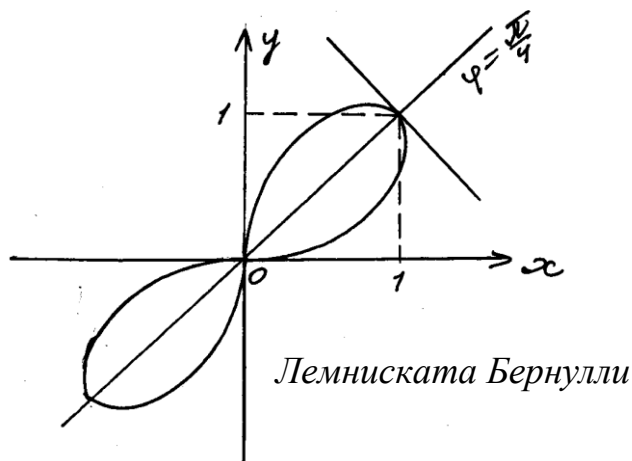


4)  $r^2 = 2 \sin 2\varphi$ ,  $\sin 2\varphi \geq 0$ ,  $\varphi \in \left[0; \frac{\pi}{2}\right] \cup \left[\pi; \frac{3\pi}{2}\right]$

$$T = \frac{2\pi}{2} = \pi$$

$$r = \sqrt{2 \sin 2\varphi}$$

| $\varphi$        | 0 | $\frac{\pi}{12}$ | $\frac{\pi}{8}$      | $\frac{\pi}{6}$      | $\frac{\pi}{4}$ | $\frac{\pi}{3}$      | $\frac{3\pi}{8}$     | $\frac{5\pi}{12}$ | $\frac{\pi}{2}$ |
|------------------|---|------------------|----------------------|----------------------|-----------------|----------------------|----------------------|-------------------|-----------------|
| $\sin 2\varphi$  | 0 | $\frac{1}{2}$    | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1               | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$     | 0               |
| $2\sin 2\varphi$ | 0 | 1                | $\sqrt{2}$           | $\sqrt{3}$           | 2               | $\sqrt{3}$           | $\sqrt{2}$           | 1                 | 0               |
| $r$              | 0 | 1                | $\sqrt[4]{2}$        | $\sqrt[4]{3}$        | $\sqrt{2}$      | $\sqrt[4]{3}$        | $\sqrt[4]{2}$        | 1                 | 0               |



$$x = r \cos \varphi = \sqrt{2 \sin 2\varphi} \cos \varphi$$

$$y = r \sin \varphi = \sqrt{2 \sin 2\varphi} \sin \varphi$$

$$x'_\varphi = \sqrt{2} \left( \frac{\cos 2\varphi}{\sqrt{\sin 2\varphi}} \cos \varphi - \sqrt{\sin 2\varphi} \sin \varphi \right)$$

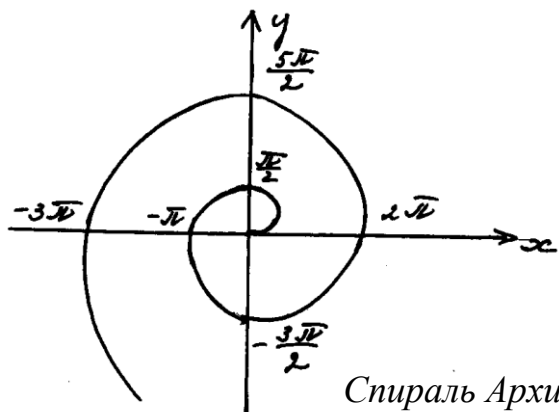
$$= \sqrt{2} \frac{\cos 2\varphi \cos \varphi - \sin 2\varphi \sin \varphi}{\sqrt{\sin 2\varphi}} = \frac{\sqrt{2}}{\sqrt{\sin 2\varphi}} \cos 3\varphi$$

$$y'_\varphi = \sqrt{2} \left( \frac{\cos 2\varphi}{\sqrt{\sin 2\varphi}} \sin \varphi + \sqrt{\sin 2\varphi} \cos \varphi \right)$$

$$= \sqrt{2} \frac{\cos 2\varphi \sin \varphi + \sin 2\varphi \cos \varphi}{\sqrt{\sin 2\varphi}} = \frac{\sqrt{2}}{\sqrt{\sin 2\varphi}} \sin 3\varphi$$

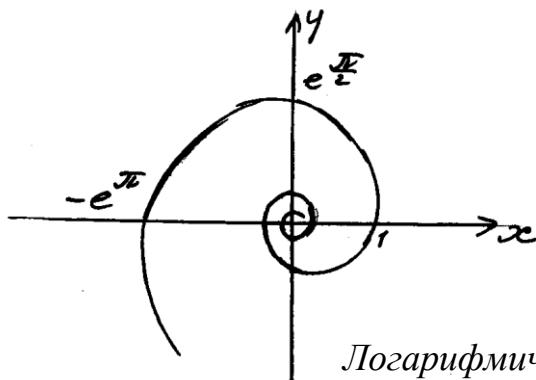
$$y'_x = \frac{y'_\varphi}{x'_\varphi} = \operatorname{tg} 3\varphi, \quad y'_x|_{\varphi=\frac{\pi}{4}} = \operatorname{tg} \frac{3\pi}{4} = -1$$

5)  $r = \varphi, 0 \leq \varphi < +\infty$



Спираль Архимеда

6)  $r = e^\varphi, -\infty < \varphi < +\infty$



Логарифмическая спираль